Understanding myCourses: A Multi-Dimensional Study of RIT's Digital Learning Environment

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This qualitative study explores the apparent shortcomings of myCourses, the learning management system used at Rochester Institute of Technology, and through thoughtful analysis discusses potential areas of improvement. Our primary contribution lies in the fact that we applied three independent qualitative methods, Survey-based Study, Ethnographic Study, User-interviews, to analyse the critical weaknesses of myCourses. This diversified approach yielded insightful feedback on various aspects, including navigation, communication tools, accessibility, and mobile device usage. This paper discusses these findings in detail and offers a foundation for future research.

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1 INTRODUCTION

E-learning platform provides digital alternatives to traditional teaching methods, with increased popularity, more and more institutions are integrating websites like myCourses as supporting tools in their teaching methods. The platform provides multiple essential features between students and faculties, such as notification, assignments, discussion and class materials. With it being an integral part of education, the usability of such platforms determines students' success in their education journey[2]. Inspired by a prior study that examined University portal[4], our goal is to examine the usability of myCourses and establish a basic understanding of the cause of users' poor experience with this platform. We will be conducting three different types of qualitative research, namely surveying, interviewing and ethnographic study. We believe a combined approach will provide more insights and findings regarding the issue. This report will focus on the evaluation of myCourses platform using qualitative approach and demonstrate the use of these three specific qualitative methods to find a scope of improving the myCourses platform.

2 METHODS

We took a multi-dimensional approach using, Survey-based Study, Ethnographic Study, User interviews, for collecting the data. For the Survey-based Study, we sent out a Google form to 10 individuals, but only 6 out of 10 participated. For the User-interviews, we conducted two supervised interviews. For the ethnographic study, we only observed one individual while giving some instructions. The demographic characteristics we examined include Age Group, Gender

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Identity, Highest Level of Education, and Employment Status. Table 1 presents a detailed demographic summary of the participants, segmented into different categories. Our participant pool spans various age groups, with a notable presence in the 25-30 age bracket, indicating a younger demographic. Gender identity among participants is also diverse, including male, female, and non-binary individuals, ensuring a broad perspective in data collection.

Demographic	Categories	Survey-based Study	Ethnographic Study	User Interviews
Age Group	15-20	1	0	2
	25-30	4	1	0
	31 or older	1	0	0
Gender Identity	Male	2	0	1
	Female	3	1	1
	Non-binary	1	0	0
Highest Level of Education	High School Graduate	1	0	2
	Bachelor's Degree	2	1	0
	Master's Degree	3	0	0
Employment Status	Employed full-time	0	0	0
	Employed part-time	2	1	0
	Student	4	0	2

Table 1. Demographic Summary with different Methodological Approaches

Survey-based Study: In this qualitative approach, we explored user experiences with myCourses through a carefully designed survey. The survey was sent out to ten participants, encompassing a diverse range of age groups, gender identities, educational backgrounds, and employment statuses. However, only six participated in giving their responses on time. Our survey comprised a series of questions aimed at assessing the usability, satisfaction, and accessibility of myCourses, as well as gathering suggestions for improvements and understanding the platform's mobile experience.

User Interviews: A classic user interview method was also used for collecting qualitative data. For the same 2 participants were found and interviewed, providing valuable insights into their experiences with using myCourses. The interviews involved a structured approach to gather information on their usage, challenges, and suggestions for improvements within the platform. Firstly, the interview process consisted of open-ended questions made specifically to elicit detailed responses from the candidates. Questions were designed to explore various aspects such as their degree program, mode of classes, grades checking process, usage of myCourses, interaction with advisors, use of additional tools such as Starfish, and specific challenges faced within the platform. The interviews were conducted in an informal yet guided manner, allowing the candidates to express their experiences freely while ensuring the conversation remained focused on extracting relevant information pertaining to their interactions with myCourses.

Ethnographic Study: Ethnographic study provides most direct findings by observing the actions executed by the user. We chose to include ethnographic study for the reason that the result is not affected by participants' memories as they will be performing tasks in their environment, therefore reducing the need to recall any details or sentiments while interacting with the website. 1 participant was recruited for the study with 5 tasks assigned to simulate the daily usage of a student.

3 RESULTS

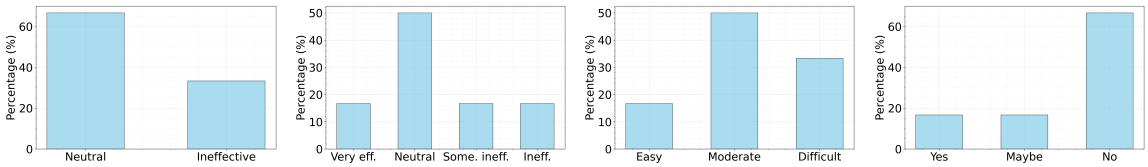
The following three sections will discuss the findings from the three different qualitative studies.

3.1 Survey-based Study

Figure 1 shows the frequency of responses for each of the different questions that were being asked. It is evedent that people were divided with their views about the ease of access for navigation and mobile experience. However, the majority of the participants indicated that myCourses is not effective for people with disabilities, while some indicating it is neutral. It is worth noting that none of the participants had any disabilities, so their views regarding this is not from personal experience.

Apart from collecting multiple answer question, we also collected user's feedback, which are summarized below.

- *User Interface and Navigation:* Users reported challenges with navigating the platform, particularly with marking discussions as read and accessing the "progress" page. Suggestions include adding a "mark as read" button for posts, improving visibility and formatting of the progress page, and enhancing the homepage layout for easier course identification.
- *Communication Tools:* Participants found the existing communication tools to lack modern touch. Participants suggested implementing a group chat option similar to Slack for each course and improving the notification system for new assignments and deadlines.
- *Email Notifications:* Users expressed frustration over the volume and lack of control over email notifications, especially regarding grade updates from tools like WebAssign. There's a call for more fine-grained control over notifications, allowing users to customize or mute certain types of alerts.
- *Grading Information:* Feedback highlighted the need for clearer grading information, including the request for grades to be presented with corresponding percentages to avoid manual calculation.
- *Accessibility and Usability Enhancements:* Several responses pointed to the need for a more intuitive assignment submission process, a clearer and more accessible progress section, and the introduction of a dark mode for the platform.



(a) Accessibility for People with Disabilities (b) Ease of Finding Information (c) Mobile User Experience (d) Willingness to Participate in Interviews

Fig. 1. Assessment of User Experiences and Accessibility of myCourses: Insights from a Survey-Based Study

3.2 User Interviews

The interview was conducted on 2 participants and the data collected was carefully analysed to identify recurring patterns, themes, and notable points from each candidate's responses. This analysis aimed to extract the key observations and insights related to myCourses usage, challenges faced, and suggestions for enhancing the platform's functionality.

Here the actual data obtained from the interviews, after some minor refinement and organization has been represented in the tabular form above Table 2. It shows the responses given by our 2 participants for various aspects/questions asked such as, checking grades, efficiency in submitting assignments and navigating through the course content,

Candidate	Moshana Green	Shane
Major	Psychology	Electrical Engineering
Age	19	19
Mode of Classes	Mostly in-person, one online asynchronous class	Offline classes for the program
Grades Check	Not challenging	Able to check grades within the platform
myCourses Usage	Visits classes, checks work to-do list, views upcoming events	Log in, access classes, review content, complete assignments
Missed Assignments	None	Few
Appointment with Advisor	Registers for classes, emphasizes open communication	Only when necessary for classes
Use of Starfish	Used for academic purposes, easily navigable	Used if needed, sometimes
Class Recording	Uses for online course, comfortable with viewing in myCourses	Uses sometimes, finds platform less intuitive
Course-specific Tools	Highlights differences in using and downloading course materials	Seeks standard format for each class, emphasizes need for reminders
Reminders for Deadlines	Lacks reminders for assignment deadlines	Mentions the lack of reminders
Suggested Feature for myCourses	Seeks feature related to reminders for deadlines	Expresses need for syncing myCourses with homework modules, uniform structure

Table 2. Comparative Analysis of Student Experiences with myCourses

etc. Both the participants are undergraduate students and use the platform daily. The features like grade checking, assignment submission and navigation were not found to be an issue for these daily users, however the described a lack of linear/symmetric organization of content and well as need of reminders. The user-centric approach in this process resulted in understanding the perspectives and experiences of the candidates to create a comprehensive overview of their interactions with myCourses. This approach facilitated the extraction of valuable qualitative data that can be utilized to inform potential improvements and enhancements to the myCourses platform. Below is the User Interview Findings after analyzing the data collected.

User Interview Findings:

Participant 1:

- *myCourses Usage:* Moshana's approach to accessing her coursework primarily involves navigating through her classes, work to-do list, and upcoming events, indicating a preference for a more organic navigation experience.
- *Use of Starfish:* Moshana finds the use of Starfish for academic purposes easy and navigable, highlighting its value as an academic support tool.
- *Suggested Feature:* Moshana suggests the implementation of a reminder feature for assignment deadlines within the myCourses platform to further enhance user experience and deadline management.

Participant 2:

- *myCourses Usage:* Shane accesses classes, reviews content, and completes assignments on myCourses, but finds the platform less intuitive, indicating a need for improved user experience.

- *Missed Assignments:* While Shane has not missed most assignments, the lack of reminders for deadlines is noted, suggesting an opportunity for enhanced deadline management within the platform.
- *Course-specific Tools:* Shane seeks a standard format for assignments and quizzes across all courses, expressing the need for consistency and streamlined user experience.
- *Syncing Courses with Homework Modules:* Shane expresses a need for syncing myCourses with homework modules and a uniform structure across courses for a more cohesive user experience.

3.3 Ethnographic Study

The participant is 27 years old, with experience using a similar system during college, but without prior experience or knowledge with myCourses. 5 tasks were given to simulate daily usage of myCourses' users. The tasks consist of navigating the system to get information, communicating with classmates/group members and reaching out to faculties. Below are the 5 tasks assigned.

- Open all the homework assignments for this week;
- Find lecture material for this week;
- Message the group member for the group project(if applicable);
- Navigate to the class recording(if applicable);
- Find updates on homework assignments and announcements for next week.

Throughout the session, there is a steep observed learning curve where the participant struggled the most in the first two tasks with most difficulty, expression of confusion and frustration. Due to the lack of search function, useful help page and consistent navigating method and course structures.

With enough exploring for the first two tasks, the participant was able to complete the latter three tasks quickly with minimal struggle. Another reason to consider is that the features required to execute the tasks are the most static functions set in specific places across courses.

After the study, a follow-up chat with the participant with questions regarding her thoughts on the system revealed, in which she replied she hated the system. Participant stated that there are too many routes to do the same thing. It is unnecessary to implement different ways to do the same thing. It would be good to build basic functions, and make them clear to use for the users, because each route looks different but also leads to the same thing. "All in all, the bottom line is, it does what it needs to do."

4 DISCUSSION

Across the findings of three studies, navigation becomes a constant theme of topic indicating underlying issues with the structure of the website. Consistency was mentioned along with navigation during the study, as the participant in ethnographic study put "there are too many was to do the same thing, they all look different". The survey consists of people with certain experience of the website, which they expressed the demand to implement more advanced functions such as group chat, and basic function that needs improvement such as reminder that includes assignment deadlines. The participant for ethnographic study had no prior experience which reproduced the experience of newly enrolled students. The most she struggled with was navigation for course materials and exhibited a steep learning curve, which soon overcame by exploring the website. Interviewing finds that participants want a better navigating experience, consistent format across all courses with a common structure to access the content, reminders for assignments and

deadlines, and a way to synchronise different modules required for assignments with myCourses to submit the work. Combining all the results, there definitely seems to be a scope for improvement in various different areas.

Although the insights collected were rich, the study is severely limited with the number of participant. Variation in experience and background with the system should be considered during the study. The lack of inclusively of the study should also be addresses for future works, as all participants are able-bodied without cognitive limitations. As previous study shown, the problems reported by participants can be amplified by users' disabilities[1]. A website built with accessibility in mind would also increase the clarity, navigating experience and ease of use to users without disabilities[3]. The study focused on the most used features of the website but overlooked under-utilized features, study of these features could be helpful to understand the design intent of the website.

5 CONCLUSIONS AND FUTURE WORK

Future work could focus on experienced participants to examine whether there are shortcuts, more systematic or intuitive approach to navigate the system with the accumulation of experience. A group ethnographic study with follow up focus group would be ideal for identifying flaws and collecting feedback. Another direction to take is to study faculties and how they utilize the features, decisions for structuring the course page. Different redesign solutions could be proposed to examine the possibility of reducing cognitive load of users while navigating for specific contents.

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